11	**(a)**		$5\sqrt{4} - 4 - 6 = 10 - 10 = 0$	**B1**	**1.1**	Command word 'verify' so attempts at solving score zero. Need to see the substitution of the given value(s) and minimal processing to obtain 0 on the RHS

Question			Answer	Marks	AO	Guidance
			$5\sqrt{9}-9-6=15-15=0$	**B1**	**1.1**	Need to see the substitution of the given value(s) and minimal processing to obtain 0 on the RHS
				[2]		
11	**(b)**		$\int_{(4)}^{(9)}\left(5x^{\frac{1}{2}}-x-6\right)\mathrm{d}x$	**M1**	**1.1**	Attempting the integration (ignore limits) and obtaining **at least two terms of the correct form** in $\alpha x^{\frac{3}{2}}+\beta x^2+\gamma x$ where $\alpha,\beta,\gamma\in\mathbb{R}$
			$\frac{10}{3}x^{\frac{3}{2}}-\frac{x^2}{2}-6x$ **oe**	**A1**	**1.1**	Fully correct integrated expression- coeffs need not be simplified, but they must be correct.
			$\left[\frac{10}{3}9^{\frac{3}{2}}-\frac{9^2}{2}-6(9)\right]-[\frac{10}{3}4^{\frac{3}{2}}-\frac{4^2}{2}-6(4)]$	**M1**	**1.1**	Must show substitution of corrects limits into their integrand of the correct form and subtract the right way - detailed reasoning required. NB – 4.5 – $\left(-\frac{16}{3}\right)$
			$-\frac{9}{2}-\left(-\frac{16}{3}\right)=\frac{5}{6}$	**A1**	**1.1**	**At least one line of working required before seeing the final answer** Accept equivalents or a decimal – awrt 0.833.
				[4]		

Question			Answer	Marks	AO	Guidance